Example 1.29

Find the value of $\sqrt{42.25}$.

Solution:

We can write this as $\sqrt{42.25} = \sqrt{\frac{4225}{100}} = \frac{\sqrt{4225}}{\sqrt{100}}$

Now, it is easy to compute the square root of the whole number 4225 by long division method as $\sqrt{4225} = 65$ and so, we now get $\sqrt{42.25} = \sqrt{\frac{4225}{100}} = \frac{\sqrt{4225}}{\sqrt{100}} = \frac{65}{10} = 6.5$

This is another way of tackling problems of square root of decimal numbers without any botheration of decimal symbol.



Try these

Using this method, find the square root of the numbers 1.2321 and 11.9025.

Example 1.30

Simplify: (i) $\sqrt{12} \times \sqrt{3}$ (ii) $\sqrt{\frac{98}{162}}$

Solution:

(i) Remembering the rule, $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

$\sqrt{12} \times \sqrt{3} = \sqrt{12 \times 3} = \sqrt{36} = \sqrt{6^2} = 6$

(ii) Remembering the rule, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$

$\sqrt{\frac{98}{162}} = \sqrt{\frac{2 \times 49}{2 \times 81}} = \sqrt{\frac{49}{81}} = \sqrt{\frac{7^2}{9^2}} = \frac{7}{9}$

Example 1.31

Simplify: (i) $\sqrt{2\frac{7}{9}}$ (ii) $\sqrt{1\frac{9}{16}}$

Solution:

(i) $\sqrt{2\frac{7}{9}} = \sqrt{\frac{25}{9}} = \sqrt{\frac{5^2}{3^2}} = \frac{5}{3} = 1\frac{2}{3}$ (ii) $\sqrt{1\frac{9}{16}} = \sqrt{\frac{25}{16}} = \sqrt{\frac{5^2}{4^2}} = \frac{5}{4} = 1\frac{1}{4}$

Remark: In the case of the (ii) problem one may be tempted to give the answer immediately as $1\frac{3}{4}$, but this is not correct since you have to convert the mixed fraction into an improper fraction and then use the rule $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$ $(b \neq 0)$

1.7.6 Approximating square roots

Can you write the given numbers $\sqrt{40}$, 6 and 7 in ascending order? Here $\sqrt{40}$ is not a square number and so we cannot determine its root easily. However, we can estimate an approximation to $\sqrt{40}$ and use it here.



Try these

Write the numbers in ascending order.

1. 4, $\sqrt{14}$, 5 2. 7, $\sqrt{65}$, 8

We know that the two closest squares surrounding 40 are 36 and 49.

Thus, $36 < 40 < 49$ which can be written as $6^2 < 40 < 7^2$.

Considering the square root, we have $6 < \sqrt{40} < 7$.

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